

6.6 Subprograms	6.6.1	be able to write programs that use pre-existing (built-in, library) and user-devised subprograms (procedures, functions)
	6.6.2	be able to write functions that may or may not take parameters but must return values, and procedures that may or may not take parameters but do not return values
	6.6.3	understand the difference between and be able to write programs that make appropriate use of global and local variables

Onscreen problem solving with programming

Students should engage with a broad range of activities to enable them to learn to program to the standard required for this qualification.

Mastering programming involves much more than simply learning the syntax/semantics of a programming language. It also involves learning strategies for problem solving, embracing mistakes as opportunities to learn, mastering a few simple tools and working together with others to achieve goals.

We have adopted Python programming language as the vehicle to explore, learn and assess the aspects of problem solving and programming covered in this qualification. Python is popular and commonly used in education.

The requirements of this qualification can be met using the Python 3 programming language. Centres are advised to choose the most up-to-date version of an interpreter that supports Python 3. There are many freely available Python 3 interpreters, often installable as part of an integrated development environment (IDE).

Programming environments

Online guided tutorials are useful in motivating and engaging students. They normally present small challenges where students copy or reproduce code to achieve a short-term goal, often based on moving up levels. Using these environments exclusively may not prepare students for the type of problems they will encounter in Paper 2. Teachers should consider highlighting how each level relates directly to the requirements of the specification. For example, if students are directing a tank to move forward until it encounters an obstacle using a while loop, then the corresponding structure of the while loop could be explored in a non-graphical text-based environment, including on paper.

Students will need experience using an IDE to write, run and debug their programming code. Developing skills in using an IDE will allow students to be more productive, especially in time-constrained activities. A suitable IDE includes these functionalities:

- an editor, with line numbers
- a syntax highlighter and checker
- breakpoints and stepping
- a variable inspector.

There are many IDEs available to freely download online. Teachers are advised to evaluate several before selecting one suitable for their students.

Classroom strategies

Several professional organisations exist for computing and computer science teachers. It is advisable to join one or more of these organisations to keep up to date and share current best practice in the teaching of computer science. A range of strategies that teachers could consider in teaching programming include:

- unplugged activities, which focus on reading and tracing code before attempting to write code
- correcting errors in code, both on paper and in an IDE

- peer instruction to encourage students to learn from each other and develop skills in reading code
- Parsons problems, where code must be arranged in order, reducing the need for copying or typing in code
- showing consistent exemplar code, using the good programming practices set out in the Programming Language Subset (PLS).

Information on any unfamiliar strategies can be found online, using a search engine.

Programming Language Subset (PLS)

Both teachers and students are encouraged to look at the Programming Language Subset (PLS). The PLS represents a specific set of constructs that are sufficient to successfully answer any question in Paper 2. Any student successfully using more esoteric or complex constructs or approaches not included in the PLS will still be awarded marks in Paper 2 if the solution is valid. The constructs contained in the PLS are found in most high-level programming languages and therefore form a foundation of computer science knowledge required in the assessment.

The PLS will be valid for the lifetime of the qualification. However, if updates are required, we will ensure the latest version is published by the 31st of January in the year of the examination. Please note that students will be given both printed and electronic versions of the PLS during the examination. Please see the *Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Computer Science – Paper 2 Instructions for the Conduct of the Examination (ICE)* for further information. The PLS and ICE are available to download on the [GCSE in Computer Science section of our website](#).

Teacher support

We will develop a range of teaching and learning materials to help centres prepare for the onscreen assessment of problem solving with programming. These include a good programming practice guide elaborating on and exemplifying the constructs in the PLS, and programming exercises with commentaries and solutions. The exercises can be used by teachers either as classroom activities or for formative assessment purposes. These teacher support materials will be made available on the [GCSE in Computer Science section of our website](#).

Practical Programming Statement (PPS)

Centres are required to complete a Practical Programming Statement (PPS), which confirms they have taken reasonable steps to ensure that each student sitting our GCSE in Computer Science has had the opportunity to demonstrate their programming skills in terms of design, write, test and refine programs in Python during their course of study. The PPS should be completed by a member of the senior leadership team at the centre. Failure by a centre to provide a completed PPS to us in a timely manner will result in potential malpractice and/or maladministration.

The PPS will be made available by the 31st of January in the year of the examination on the [GCSE in Computer Science section of our website](#).

The completed PPS must be submitted to us by the 31st of May in the year of the examination. Centres are reminded that if they do not submit this form it may result in malpractice/maladministration.

Instructions for the conduct of the examination (ICE)

For detailed information on the set up of candidate secure user areas, download of coding files, administration of the examination and transmission of candidates' completed coding

files to us for marking, please refer to the *Pearson Edexcel Level 1/Level 2 GCSE (9-1) in Computer Science – Paper 2 Instructions for the Conduct of the Examination (ICE)*.

This document will be updated on an annual basis so that in each year of examination the version to be used will be available by the 31st of January. It is available on the [GCSE in Computer Science section of our website](#).

Security and backups

For materials stored electronically, centres are strongly advised to utilise firewall protection and virus-checking software, and to employ an effective backup strategy, so that an up-to-date archive of students' evidence is ensured following the examination.